

1 **How Digitalization Unleashes Endowment Advantages for Green**

2 **Innovation: Evidence from China’s county-level panel data**

3 **Abstract**

4 Sufficient supply of green technologies is the vital path for the sustainable development of
5 ecological conservation areas. Although these areas have abundant ecological endowment, their
6 comparative advantages are suppressed due to mismatch between endowment structure and
7 prevailing innovation paradigm. On the basis of innovation system theory, this study incorporates
8 the endogenous structural evolution perspective from new structural economics to examine how
9 digitalization alleviates this mismatch. Using panel data of China’s ecological function zones (EFZs)
10 from 2012—2023, empirical results have shown a mutual convergence dynamic: digitalization
11 effectively accelerates the upgrading of local endowment structure to meet the traditional innovation
12 paradigm; it also alters the innovation supply paradigm by activating digital-green synergy and
13 stimulating inter-regional innovation market network. A proactive government could amplify this
14 effect through digital-powered environmental regulation. The heterogeneity findings indicate that
15 the impact of digitalization on green technological innovation is related not only to socioeconomic
16 factors but also to the abundance of local ecological endowment, which shapes the digital
17 application scenarios. These findings highlight the broader effect of digitalization on long-term
18 economic evolution and innovation paradigm shifts, offering insights for latecomer economies
19 pursuing green development.

20 **Keywords:** Green productivity, digitalization, green innovation, ecological function zones,
21 ecological endowment

22 **JEL:** O13, O30, O44

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1. Introduction

The supply of green technologies is a key constraint on the development of a modern ecological economy, and green technological innovation (GTI) has become the most important support for the development of characteristic ecological industries and their large-scale operation in areas with ecological endowment advantages (Liu et al., 2024; Wen et al., 2025). From the perspective of knowledge economic history, after institutional innovations in the pre-Industrial Revolution era, public technology and knowledge supply were what modern growth relied on (Mokyr, 2005). GTI is thus the foundation of sustainable growth, and it provides essential devices and facilitates more effective management schemes for both regional environment governance and global climate crisis treatment (Irfan et al., 2025). The GTI is therefore a main path toward socioeconomic green transition. GTI means even more in regard to less-developed rural and other eco-constrained areas, which face structural dilemmas between ecological conservation and economic development, industrial resource scarcity and ecological endowment abundance (Shen et al., 2025). Proper utilization of ecological resources and simultaneous realization of ecological service value are possible only via sufficient green technology supply. It is not only a crucial path for growth in these areas but also a significant measure for obtaining a balanced regional development pattern.

Although green technological innovation has many sustainable dividends, the current innovation system and incentive mechanism pose numerous challenges. Large prerequisite investments limit GTI to a few leading enterprises and institutions funded by wealthy local governments, leaving the majority of MSEs and peripheral regions behind (Losacker et al., 2025). Trade-driven knowledge spillover and technology diffusion benefit mainly node cities, expanding regional gaps even further. Focusing on ecological zones, this dynamic becomes more complex. The traditional technology supply paradigm is one of the main barriers in such a context. In this respect, experience from China's National Key Ecological Function Zones (EFZs) is valuable. EFZs are county-level areas with regional or national ecological significance. Conservation is therefore the primary principle for these areas. However, another perspective of the characteristics of EFZs is their abundance of ecological resources, which provides unique opportunities to develop compatible eco-friendly industries. Accordingly, special funds and transfer payments from governments are provided to support ecological conservation and green development in EFZs. More policy details are presented in Figure 1.

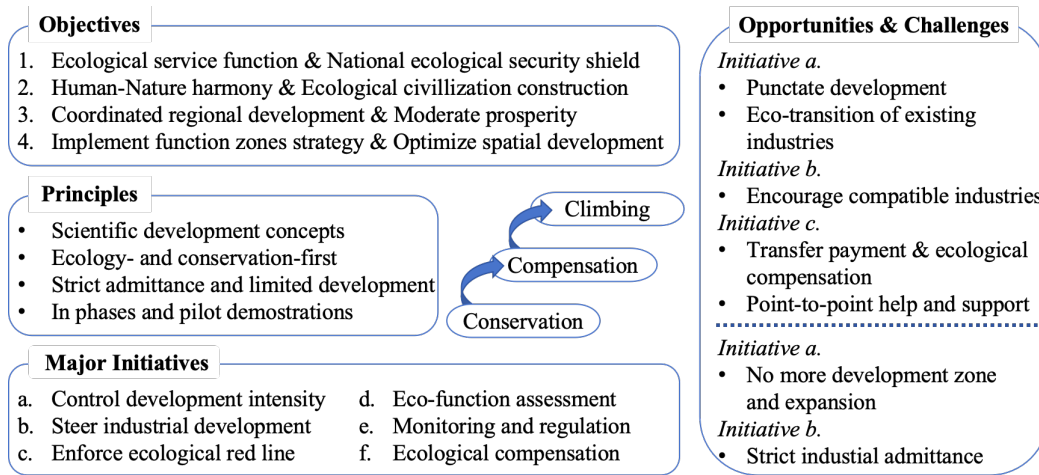


Figure 1. Policy overview

The EFZs require sufficient green technologies to establish compatible ecological industries and achieve ecological value realization. Moreover, EFZs should have a stronger influence on GTI according to the new structural economics idea of factor endowment. This dynamic constitutes a powerful green technology market, yet in reality, EFZs have been weak in GTI since their origin. As shown in Figure 2, the number of acquired green patents in EFZs is always much smaller than that in other counties and districts. Behind this endowment-disobeying phenomenon is the urgency of rearranging the technology supply paradigm in these regions. The traditional technology supply paradigm was developed under industrialism and was built for standardized mass production. However, the utilization of ecological resources under conservative constraints requires flexible and fast-responding methods for small-lot customized production. This mismatch represents the main barrier to the path of EFZs to innovative and sustainable development.

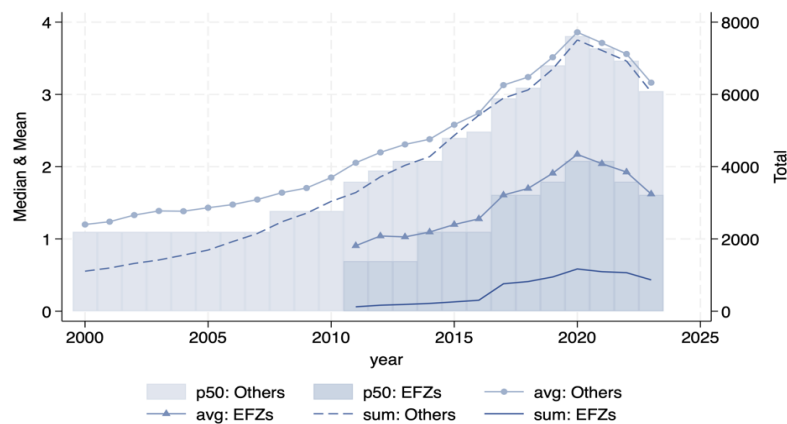


Figure 2. Differences in green innovation between EFZs and other regions

The need for a new technology supply paradigm requires efficient information adaptation and flexible production arrangement mechanisms, which aligns with digitalization. Socioeconomic

digital transition provides theoretical potential to address the traditional technology supply paradigm barrier of EFZs. According to the prevailing literature, digitalization enhances the communication efficiency of supply chains (Fan et al., 2024). It helps optimize enterprise resource accumulation, utilization and distribution (Shao et al., 2024). Digitalization is also crucial for achieving fast-response in the face of a green transition (Wang et al., 2025). Returning to the classic literature from Mokyr (2005), lowering knowledge acquisition cost is the premise of persistent technological innovation. The social nature of knowledge is crucial to realizing its economic value. Correspondingly, the network effect of digitalization benefits both acting mechanisms. These factors could change the existing innovation path, providing opportunities for EFZs to establish effective green innovation paradigms. However, this nexus has not been thoroughly examined from the perspective of technology supply.

This research aims to identify the systematic effects and structural mechanisms of digitalization on GTI in eco-constrained rural areas to provide insights into digital–green synergy development. On the basis of a balanced county-level dataset covering 523 EFZs in China from 2012--2023, this paper constructs a comprehensive index system to evaluate digitalization and then adopts the TWFE-OLS model to estimate the elasticity of GTI with respect to digitalization. A series of theoretically linked robustness checks, including the ZINB model, DPD model and quasi-natural experiment, are conducted to validate causal inference. Two mechanisms of technology supply are examined via causal mediation models. Quantile regressions on GTI and the policy interaction effect are investigated to explore more impact factors and a more sophisticated mechanism. Finally, this paper uses FMM combined with interaction models to identify heterogeneous paths caused by local ecological endowments.

This work reveals that digitalization significantly promotes GTI in EFZs and that two aspects of the technology supply paradigm serve as powerful mechanisms. Although this effect diminishes as GTI advances, it remains positive for all EFZs. Digitalization and environmental regulation strengthen each other. Local ecological endowment induces heterogeneous effects of digitalization with respect to EFZ types. This study reassessed county-level digitalization and applied new structural economics to discuss the normative and empirical nexus between local digital capabilities and GTI networks within EFZs. This study extends the innovation system theory by incorporating the perspective of structural evolution from new structural economics (NSE). This integration gives

rise to the concept of innovation paradigm, through which the broader effects of digitalization on innovation activities and structural mismatch could be discussed. This study aims to connect and advance the theoretical understanding of digitalization and innovation systems, providing insights for digital-green synergy in less-developed rural and other eco-constrained areas. The conceptual framework is illustrated in Figure 3.

The remaining sections are arranged as follows. Section 2 presents the literature review. Section 3 presents the theoretical derivation and proposes hypotheses. Section 4 introduces the research design, variables used and model specification. Section 5 presents and analyzes the empirical results. Section 6 provides conclusion-based policy implications and prospects.

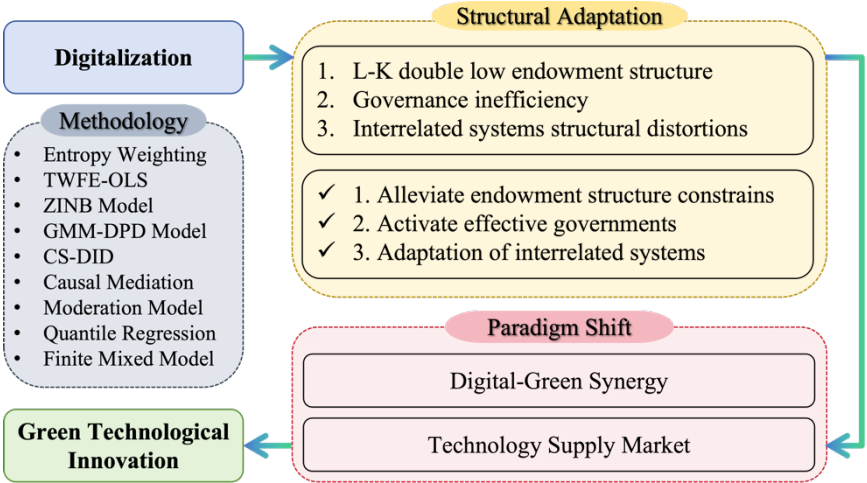


Figure 3. Conceptual framework

2. Research background and literature review

2.1. The structural context of EFZs

Regarding the relationship between digitalization and green innovation, researches on EFZs are promising to provide a new endowment structure constraining context. When it comes to economic structures, NSE serves as an insightful theoretical tool. The NSE carbon neutrality theory (Lin et al., 2025) argues that while the structures of factor endowments, energy endowments, and ecological endowments in an economy evolve endogenously, they determine the production, living, and ecological (PLE) systems at the next level. Moreover, through six interrelated systems including finance, space, infrastructure, and institutions, etc., they shape distinct economic landscapes. A proactive government and an effective market play active and crucial roles in regulation and coordination during this process. Figure 4 provides a simple sketch map illustrating this theoretical framework.

The EFZs possess green-biased factor endowments and ecological endowment structures characterized by abundant ecological resources. This policy not only can effectively control the conversion of ecological land to non-ecological land and increase the supply of local ecological service value (Chen & Yao, 2023) but also facilitate regional access to vertical ecological compensation transfer payments, thereby improving internal green economic efficiency (Yang, 2023). Additionally, it can promote the coordinated development of ecological and economic quality within the region by driving the spatial agglomeration of green ecological industries, exerting economies of scale, and optimizing the industrial structure (Shen et al., 2025). Therefore, EFZs should inherently have significant comparative advantages in green production, green living, and ecological development, and GTI that aligns with these comparative advantages should also have stronger viability and competitiveness.

The backwardness of GTI in EFZs in the real economy is due to structural distortions, as shown in Figure 4. These include the lack of innovation infrastructure (Li et al., 2024; Huang et al., 2025), brain drain caused by underdeveloped local higher education (Alfalih & Hadj, 2024), financing constraints resulting from weak rural finance (Li et al., 2024), excessive regulation due to backward institutional design (Sun et al., 2024), and misallocation of subsidies (Song et al., 2022; Wang et al., 2024). Furthermore, most EFZs are economically underdeveloped areas with a net population outflow, and thus have a low L-K factor endowment structure. As a result, despite their rich ecological resources, the insufficiency of complementary factors greatly restricts GTI. Therefore, treating structure distortions is crucial to crack the GTI dilemma in EFZs.

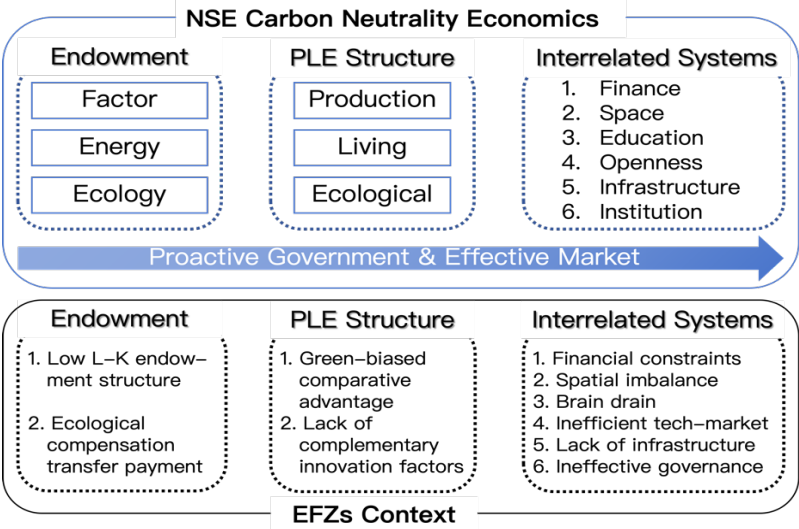


Figure 4. The NSE perspective & the EFZ context

2.2. Literature on GTI and digitalization

The role of GTI in reducing carbon emissions is well established (Irfan et al., 2025). It is advanced by financial sector innovations (Zhou et al., 2025), and its interaction with financial depth significantly decreases the carbon footprint (Ramzan et al., 2023). Core technological advances further enhance carbon stock disposal and foster eco-friendly economies (Zhou et al., 2024). A recent study based on the EU market revealed that GTI can substantially increase the shadow price of carbon emissions, thereby better transmitting ecological protection constraints to agents as incentives (Zhu et al., 2025). Moreover, under energy constraints, GTI can maintain or even enhance corporate profitability (Jia et al., 2022). More divergent research even suggests that holding green technology patents enhances the pricing efficiency of financial tools such as credit default swaps (CDS) (Rahman, 2024). Overall, GTI exerts multilevel and multiagent impacts on the green transition, making accelerating its implementation crucial for carbon-free economic growth, particularly in later-developing regions such as EFZs.

In the real economy, GTI faces complex challenges. The first layer is the regional context. In regions with greater specialization and stronger R&D capabilities, firms exhibit a stronger preference for locally developed green technologies (Losacker et al., 2025), which indicates that GTI outcomes from advanced regions face challenges in diffusing to other areas. GTI in lagging regions may even be suppressed to some extent due to talent and industrial siphoning effects (Luo et al., 2023). Furthermore, a study based on 401 NUTS-3 regions in Germany revealed that both start-ups and mature firms exhibit strong trajectory persistence and path dependence in their green innovations (Hansmeier & Losacker, 2024). This, suggesting that regions lagging in GTI find it difficult to achieve breakthrough and maintain diversity in GTI pathways. In some underdeveloped regions, financing constraints and insufficient investment often constitute major barriers to local GTI, where digital financing can provide assistance (Shao & Chen, 2023). With respect to governance, the profound impact of government attention on GTI is emphasized (Yang et al., 2024), and the mechanism of attention competition is revisited.

Digitalization presents multifaceted solutions for these barriers of GTI. Digitalization strengthens supply chain communication and efficiency (Fan et al., 2024), whereas green digital finance effectively advances technology diffusion (Tan et al., 2024). Further research reveals that corporate digitalization enhances firms' capacity to absorb, utilize, and monetize green innovations (Ning et

al., 2023) while improving their efficiency in terms of resource accumulation, utilization, and allocation (Shao et al., 2024), thereby strengthening microlevel innovation efficacy for regional economic sustainability. The digital economy fosters new market structures that rapidly channel innovation resources while reducing overall credit costs to stimulate GTI (Liu et al., 2023). Moreover, digitalization provides an adaptive context for GTI by enhancing environmental information disclosure and reducing environmental uncertainty (Han et al., 2024).

From a corporate network perspective, the digital integration of supply chains positively impacts both digital and green innovation in the early stages (Dong et al., 2023). Microeconomic digital transformation also drives GTI by facilitating knowledge sharing and information flow (Sun, 2024). Conversely, under green transition shocks such as green taxes, digital technologies can respond more rapidly and exert a moderating effect within the logic chain linking green transition and high-quality development (Wang et al., 2025).

In conclusion, GTI plays a vital role in advancing eco-friendly economic development, yet its implementation—especially within EFZs—faces challenges such as diffusion barriers, financing constraints, and insufficient policy attention. County-level digitalization offers a promising pathway to address these issues by enabling technology diffusion, alleviating financing constraints, and enhancing governance participation. While the literature acknowledges the potential of digitalization to strengthen GTI, gaps remain in comprehensively measuring digital transformation at the county level, accounting for the unique endowments of EFZs, and examining the underlying mechanisms from a technology supply–demand perspective. To address these gaps, this study develops a multidimensional indicator system to assess county-level digitalization and applies the principles of new structural economics to analyze how digital capabilities can activate GTI networks within EFZs. By linking digitalization with GTI through the lens of technology market dynamics, this research aims to advance the theoretical understanding of digital–green synergy.

3. Theoretical derivation

3.1. Innovation supply paradigms and the mutual convergence

The innovation system theory is a systemism interpretation of general innovation activities consisting of actors, networks, institutions, and technological features (Van Alphen et al., 2010). This theory originates from Christopher Freeman’s (1989) investigation on Japan’s technology policies, which introduced seminal concepts of national innovation system (NIS) and regional

innovation system (RIS), thereby establishing the notion of multi-layered nested innovation networks. Subsequently, this theory was further developed by scholars such as Malerba (2002), who progressively expanded the boundaries of such systems to incorporate sectoral factors and began to recognize the heterogeneous patterns through which different sectors and their interactions shape innovation, thus initially giving shape to the idea of innovation ecosystems. Later, the concept of “open innovation” further abstracted the knowledge flows within innovation networks and the subsequent marketisation of innovation outputs (Chesbrough, 2019; Chesbrough et al., 2006). Like many other system theories, innovation system theory has faced criticism regarding its vague boundary. In recent years, several Nordic scholars have sought to formulate more clearly delimited definitions via literature reviews and philosophical modelling (Skjølsvik & Kaloudis, 2023). For instance, Andersson et al. (2023) put three propositions to clarify the technological innovation system (TIS) concept:

Proposition 1: “A TIS is a set of structures and processes that are unified by their role in developing and shaping a specific technology.”

Proposition 2: “The specific technology, which is developed and shaped by a technological innovation system, can be understood as a production-consumption system.”

Proposition 3: “The structure of a TIS, as well as the associated production-consumption system, is social and technical, and may also include ecological elements.”

Following this theoretical trajectory, this study introduces the endogenous evolution perspective from NSE, and defines innovation paradigm as the temporal configuration of the innovation system under certain innovation input set and endowment structure. The innovation paradigm is jointly shaped by the feasibility conditions offered by the endowment structure and the sufficient conditions required by current marginal technological characteristics; it represents a snapshot of the evolving innovation system at a certain time point. The innovation paradigm could be interpreted as “how we conduct innovation.” Unlike evolutionary economics, where innovation is bound by firm “routines” (Nelson & Winter, 2004), the NSE perspective emphasizes the transformability of the endowment structure-rooted innovation paradigm. It evolves with changes in the endowment structure and technological features. In an economy with proactive government and efficient market, this evolution should be synchronous; distortions in the interrelated systems may induce misalignment. The innovation supply paradigm is thus a production-consumption understanding of the innovation system, with a focus on how technological products are generated and how they are marketized.

Knowledge spillover and technology diffusion are the key mechanisms on the input and output sides of the innovation paradigm; their direction, intensity, and content evolve endogenously with the paradigm itself, shaped by changes in the underlying endowment structure.

The disadvantage of EFZs and other regions in GTI can therefore be explained as a mismatch between the feasibility conditions afforded by local endowment structure and the current innovation paradigm. Specifically, the low L-K endowment structure of EFZs fails to meet the demands of the prevailing capital-intensive innovation paradigm (Jiang et al., 2023; Hölzl & Peneder, 2019). Consequently, a fairly intuitive potential solution lies in promoting the evolution of EFZs' factor endowment structure to bring it closer to the innovation paradigm, or alternatively, in advancing a shift of the innovation paradigm to align it with the endowment structure that EFZs can offer. We term this process “mutual convergence”, as shown in Figure 5. Extensive literature has amply demonstrated that digitalization holds significant theoretical potential at both ends of this mutual convergence process.

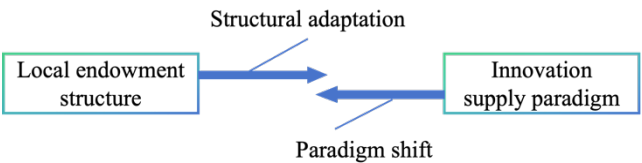


Figure 5. The mutual convergence process

3.2. Network effect and structural adaptation

Digitalization exerts a widely-documented positive effect on innovation. However, for EFZs, it more specifically alleviates the mismatch between their low L-K endowment structure and the prevailing capital-intensive innovation paradigm. This is a theoretical prediction that the NSE proposes beyond classical innovation system theories and evolutionary economics.

First, digitalization could improve the low L-K endowment structure of EFZs via network effect. The network effects of digitalization break down the spatial barriers to the flow of innovative factors, enabling EFZs to participate in GTI by leveraging their comparative advantages in ecological resources, without completely relying on locally scarce capital and high-skilled labor. Ultimately, this not only allows EFZs to leverage their unique strengths in GTI but also helps them realize greater market value through localized and distinctive GTI initiatives. In accordance with Metcalfe's Law, as EFZs deepen integration into digital markets and innovation collaboration networks, the economic returns from localized green patents increase (Ma et al., 2023), accelerating the

accumulation of capital, knowledge, and data. This process constitutes a digitally facilitated structural upgrading towards the mainstream GTI paradigm.

Second, digitalization enhances the capacity of local governments to mitigate governance distortions that compound the structural mismatch. Within the NSE framework, government plays a crucial role in facilitating structural transformation (Lin, 2011). Digitalization reduces information asymmetry (Fan et al., 2024; Han et al., 2024), enabling more efficient matching between GTI projects and targeted transfer payments, and ensuring public resources flow to viable innovation initiatives rather than politically motivated “face projects”. Moreover, digital government construction enhances green total factor productivity (Chen et al., 2025) and green innovation (Yang et al., 2024). Therefore, digitalization mitigates governance distortions to promote GTI.

Third, digitalization enhances the self-sustaining capacity of innovation sectors through interrelated systems. Digitalization improves necessary infrastructure to enhance technological innovation (Zhou et al., 2024), digital finance promotes regional green innovation by alleviating financing constraints and increasing R&D investment (Shao & Chen, 2023), digital economy also helps mitigate regional imbalance in green innovation (Liu & Yang, 2025). All in all, Digital integration in EFZs can thus enhance the self-sustaining capacity of innovation sectors through both internal and external channels. Therefore, we propose Hypothesis 1:

Hypothesis 1. Ceteris paribus, digitalization facilitates regional green technological innovation.

3.3. Innovation supply paradigm shift

In addition to structural adaptation within EFZs, digitalization is also changing the traditional paradigm of technology supply. The NSE argues that the optimal production structure is rooted in the factor endowment structure of each economy (Lin, 2011; Xu, 2026). Only industries that conform to the latent comparative advantages behind the given endowment structure, could have spontaneous viability without subsidies or interventions in other forms (Lin & Tan, 1999; Lin & Xu, 2018). Therefore, it is necessary to seek a technology supply paradigm that adapts to the low L-K endowment structure. We also see an entry point in the digital transformation: the natural digital-green synergy means that under the background of digitalization, green innovation can be achieved without too much additional investment; the optimization of the technology supply market makes the cross-regional sharing of innovation factors and the circulation of technical products more

efficient.

3.3.1. Digital–green synergy

Economic digitalization can catalyze the emergence and popularization of new institutional frameworks, and green transformation also benefits from it. This organic combination is referred to as digital–green synergy. From the perspective of institutional innovation theory, institutions can be seen as specialized goods, which are subject to the demand–supply mechanism(Jiang et al., 2024). On the one hand, the digitalization process challenges traditional economic institutions, particularly intellectual property rights, exposing their limitations in addressing the production and transaction of digital network-based intellectual property. This, in turn, spurs demand for more adaptive institutional arrangements. On the other hand, the virtual market environment created by digitalization reduces institutional innovation costs and enhances the efficiency of new institutional supply through economies of scale and increased transparency. Taking intellectual property protection as an example, the application of digital technologies and centralized network platforms helps lower the costs of intellectual property regulation and protection. Consequently, digitalization drives institutional self-renewal and optimization, particularly advancing the refinement of intellectual property protection mechanisms. This fosters a more favorable institutional environment for green technological innovation (Fang & Li, 2024), ultimately supporting the green transformation of regional economies.

Hypothesis 2. Digitalization enhances digital–green synergy to promote regional GTI.

3.3.2. The supply market of technologies

Owing to the strong regional and sectoral heterogeneity in demand for green technologies, one of the primary barriers to cross-regional GTI cooperation and transfer is the internalization impetus of firms. Digitalization can effectively mitigate this issue, thereby promoting technological cooperation and diffusion. When markets are imperfect, firms often opt to allocate innovation resources and transfer technologies through means such as establishing subsidiaries. This essentially aims to substitute markets with firms to reduce transaction costs in technology supply and mitigate intellectual property risk. Digitalization helps reduce information asymmetry, lower coordination costs, mitigate market risks, and decrease factor search costs (Wang & Li, 2024), thereby easing internalization pressures and promoting technology transfer and external collaboration. Furthermore, due to the presence of network externalities, digital technology supply networks exhibit an

endogenous tendency toward self-expansion (Shy, 2006). This facilitates connections among more innovation actors, further activating inter-regional technology supply markets.

Hypothesis 3. The network effect of digitalization empowers regional GTI capacity by activating an inter-regional tech-supply market.

4. Research design

4.1. Model construction

4.1.1. Baseline model

This paper constructs a two-way fixed effect OLS logarithmic model as follows, controlling for covariates and year and county characteristics to obtain pure effect estimation.

$$\ln Gtech_{it} = \alpha_0 + \beta \ln DI_{i,t-1} + \gamma' X_{it} + \delta_i + \lambda_t + \varepsilon_{it} \quad (1)$$

where $\ln Gtech_{it}$ denotes regional green innovation capacity measured by the number of green patents acquired each year, which is the dependent variable of this research. α_0 is a constant item; given that a certain amount of time is essential for technological innovation and patent application, the first-order lag of digitalization ($\ln DI_{i,t-1}$) is more appropriate for evaluating the effect of digitalization on long-term innovation activities. β is the coefficient of interest and denotes the marginal effect of digitalization on regional green innovation capability, the statistical significance of which determines whether Hypothesis 1 is valid.

Although log-linear models in innovation-related economic research is already an established academic convention and modeling tradition (Crepon et al., 1998; Griliches, 1979), given that the DI values in the data used in this paper are close to 0, also $Gtech$ contains 0, it is necessary to assess the necessity and applicability of log-linear models in this context. We compared the explanatory power of a linear model and a log-linear model and examine the binscatter plots to verify the optimal specification. As shown in Appendix A, the empirical results suggest a log-linear model.

The matrix X_{it} represents a set of control variables, including industrial structure (IS), economic development (GDP), government intervention capacity ($Govex$), financial vitality (Fin), the student scale (Edu), and living conditions ($Elec$) (Shen et al., 2025; Wang et al., 2024). Among these factors, industrial structure and economic scale directly influence a region's motivation and potential for green digital innovation (Destek & Sarkodie, 2019; Sharma et al., 2022); government intervention capacity controls the potential impact of environmental regulation on green innovation and reflects government-led characteristics (López et al., 2011; Yang, 2023); financial vitality represents the

credit funding support available for technological innovation in the region (Ahmad et al., 2021); the number of secondary school students indicates future human capital that impacts long-term innovation (Ni et al., 2023); and since people tend to raise children in communities with better living conditions, this indicator also partially reflects residential quality. Total electricity consumption provides a direct measure of the industrial and service sector production scale, is directly linked to residents' living standards, and reflects the local environment for green technological innovation (Li & Yang, 2025).

4.1.2. Heterogeneity model

Regular grouped regression heterogeneity models may render coefficients incomparable due to sample size inconsistencies and regression estimation issues, and the significance of heterogeneity also cannot be directly identified. Interaction heterogeneity models offer a superior solution to this problem if the covariates consistency premise is valid. The control variables selected for the benchmark model in this study represent only the background factors of green innovation, demonstrating theoretical consistency. These covariables themselves do not exhibit excessive heterogeneity, making them suitable for interaction heterogeneity models. The interaction models avoid the sample selection bias inherent in grouped regression and the increased estimation errors caused by reduced sample sizes within groups. Moreover, they enable direct identification of difference significance and allow simultaneous comparisons among multiple groups, representing a more efficient testing method. The model is constructed as follows:

$$\ln Gtech_{it} = \alpha_0 + \beta_1 \ln DI_{i,t-1} + \beta_2 D_{it}^n + \beta_3 (\ln DI_{i,t-1} \cdot D_{it}^n) + \gamma' X_{it} + \delta_i + \lambda_t + \varepsilon_{it} \quad (2)$$

where D_{it}^n stands for dummy variables of EFZ types. When the sample belongs to this EFZ category, $D_{it}^n = 1$, the model is thus equivalent to:

$$\ln Gtech_{it} = (\alpha_0 + \beta_2) + (\beta_1 + \beta_3) \ln DI_{i,t-1} + \gamma' X_{it} + \delta_i + \lambda_t + \varepsilon_{it} \quad (3)$$

Where the former interaction term coefficient β_3 denotes the "additional effect" of samples of this EFZ type relative to other types, whereas $\beta_1 + \beta_3$ is the total marginal effect. Therefore, if β_3 is significant, this type of sample may exhibit heterogeneity.

4.2. Variable definition

4.2.1. Regional digital index

This paper establishes a multidimensional indicator system to measure the digitalization level of various ecological functional zones. The specific indicators used are presented in Table 1. This

framework encompasses three dimensions: industrial digitalization, digital industrialization, and digital infrastructure. Owing to significant gaps in county-level data, a small portion of indicators were derived from city-level data segmentation. To ensure that the composite indicators objectively reflect data realities, the entropy weighting method was employed for the final synthesis. The information entropy weights for each secondary indicator are detailed in the table 1.

Table 1. Regional digital index

Indicator	Sub-indicator	Weight
Industrial digitalization	Digital inclusive finance	0.009226
	Rural e-commerce pilot	0.028299
	Cross-border e-commerce pilot	0.130951
Digital industrialization	Total volume of postal and telecommunications services	0.263846
	Proportion of employees in information transmission, computer services, and software industries	0.055277
Digital infrastructure	Number of fixed-line telephone subscribers	0.074903
	Number of mobile phone subscribers	0.108377
	Number of broadband internet access users	0.084741
	Digital village pilot	0.244380

4.2.2. Digital-green coordination

To evaluate the synergy between green technological innovation and digital innovation activities, this paper employs a coupling coordination degree model (CCDM) containing two subsystems: key digital technological innovation ($Dtech_{it}$) and green technological innovation ($Gtech_{it}$). Key digital technologies are identified according to the official standard issued by the China National Intellectual Property Administration¹. To prevent value shifts caused by excessively large T values, sub-systems $Gtech_{it}$ and $Dtech_{it}$ are min-max normalized first, and the weights for both subsystems are uniformly set to 0.5, which is in strict accordance with the common practice in prevailing literature (Fan et al., 2019; Han et al., 2023; Wang et al., 2019; Xing et al., 2019). Specifically, equal weights implicitly assumes that each subsystem is equally important in the synergistic evolution, which is critical to focus the indicator on the coordination between subsystems (Yang et al., 2020; Yin & Xu, 2022). Au contraire, differentiated weighting could obscure the clarity of coupling coordination. Also, in the absence of strong a-priori evidences, assigning equal weights helps minimize potential subjective bias. The T value in the CCDM represents the general development of subsystems, whereas the C value indicates the balance between subsystems. Together, the

¹ <https://www.cnipa.gov.cn/attach/0/d32119ae1faa4fbf9e308824862f87dd.pdf>

geometrical average of T and C reflects the coordinated development of green and digital technological innovation.

$$T_{it} = \frac{1}{2}Gtech_{it} + \frac{1}{2}Dtech_{it} \quad (4)$$

$$C_{it} = \frac{(Gtech_{it} \cdot Dtech_{it})^{\frac{1}{2}}}{\frac{1}{2}(Gtech_{it} + Dtech_{it})} \quad (5)$$

$$GDC_{it} = \sqrt{T_{it} \cdot C_{it}} \quad (6)$$

4.2.3. Ecological endowment

Two adjoint variables are adopted to control the heterogeneity of the ecological endowment in the FMM estimation: the ecological service value (ESV) and the normalized difference vegetation index (NDVI). The calculation method for ESV follows the approach outlined by Zhang et al. (2023), which primarily encompasses the economic value of ecological products on the basis of equivalent factor tables, grain yields, and local historical purchase prices. The ESV index effectively reflects the abundance of local ecological resources available for economic conversion, whereas the NDVI supplements monitoring through objective satellite remote sensing technology. The NDVI evaluates vegetation coverage by analyzing differences in infrared reflectance bands. It provided a visual representation of the relative distributions of trees, grasslands, water bodies, bare rock surfaces, and desert areas. The integration of the ESV index and the NDVI allows for a more comprehensive assessment of regional ecological endowment from both economic value and natural observation perspectives. A detailed definition of the main variables is given in Table 2.

Table 2. Variable definitions

Symbol	Variable	Definition
<i>Gtech</i>	Green innovation	Total number of actually acquired green patents including inventions and utilities
<i>DI</i>	County digitalization index	Comprehensive digitalization index constructed as in Table 1
<i>IS</i>	Industrial structure	Ratio of sector value added in tertiary to primary
<i>GDP</i>	Economic development	Per capita GDP
<i>Govex</i>	Government intervention capacity	General public budget expenditure of local governments as a share of GDP
<i>Fin</i>	Financial vitality	Balance of financial institution loans as a share of GDP
<i>Edu</i>	Human capital and livability	Number of general secondary schools enrolled students per 10,000 people
<i>Elec</i>	Living standard & production	Total electricity consumption by the whole society
<i>DGS</i>	Digital-green synergy	Coupled coordination degree of green and digital innovation
<i>CTC</i>	Cross-regional green technology cooperation	Number of green patents acquired through bilateral contribution

<i>Cite</i>	Cross-regional technology market	Number of inter-county patent citations
<i>ER</i>	Environmental regulation enforcement	Number of environmental penalty cases
<i>ESV</i>	Ecological endowment	Ecological service value
<i>NDVI</i>	Vegetational cover	Normalized difference vegetation index

4.3. Data sources and descriptive statistics

Most of the data used in this research are publicly available in the statistical yearbooks of counties, districts and cities. Patent data are from the IPAT database. Interregional patent collaboration data and county-level environmental penalty case data are from the RStata Data Center, a commercial data organizing and management platform. The ecological service value is calculated with reference to the equivalent factor method developed by Xie et al. (2015).

Only prudent curation is conducted on the raw data. After a small number of outliers were removed, the missing variables were imputed via random forest regression interpolation. Given the adoption of a log-linear model, the natural logarithm was taken for all continuous variables. The final descriptive statistics are presented in Table 3. The values of each variable appear normal, which is consistent with the data characteristics and economic realities. Furthermore, the numerical differences between variables are not excessive, making them suitable for further analysis. **Notably, the minimum of *DI* is close to 0, which could lead to data dispersion. Appendix A provides detailed discussions and evaluation of such problem, the empirical conclusion suggestion that the natural-log transformation is optimal, and data dispersion does not affect estimations significantly.**

Table 3. Descriptive statistics

VarName	Obs	Mean	SD	Min	Max
<i>lnGtech</i>	5753	1.604	0.866	0.000	5.142
<i>lnDI</i>	5753	-2.363	1.128	-6.109	-0.587
<i>lnIS</i>	5753	0.731	0.597	-1.726	3.113
<i>lnGDP</i>	5753	10.247	0.553	8.408	12.984
<i>lnGovex</i>	5753	-0.913	0.691	-5.207	2.449
<i>lnFin</i>	5753	-0.357	0.594	-8.197	1.645
<i>lnEdu</i>	5753	6.040	0.537	0.693	8.118
<i>lnElec</i>	5753	11.351	1.079	5.011	15.936
<i>lnDGS</i>	5753	-2.147	0.965	-6.908	-0.003
<i>lnCTC</i>	5753	0.298	0.681	0.000	6.960
<i>lnCite</i>	5753	3.790	1.813	0.000	8.219
<i>lnER</i>	5753	0.362	0.820	0.000	5.670
<i>lnESV</i>	5753	16.080	1.335	0.000	19.650
<i>lnNDVI</i>	5753	-0.386	0.426	-2.557	-0.056

5. Empirical results and analysis

5.1. Baseline regression

Since panel data contains a time dimension, serial correlation issues may likewise arise. The Wooldridge autocorrelation test for panel data confirms the presence of first-order autocorrelation (H_0 : no first order autocorrelation, $F(1, 522) = 39.895$, $p = 0.000$). To address this issue, this paper employs robust standard errors clustered at the county level for every regression to account for within-group autocorrelation, thereby obtaining more reliable estimates of statistical significance. The fixed-effects regression based on panel data initially yielded a positive and significant coefficient. As shown in Table 4, the coefficient maintained its positive sign after controlling for variables and fixed effects. Ultimately, the TWFE-OLS model estimates a coefficient of approximately 2.683 for green technological innovation at the digitalization level of the previous period, which remains statistically significant at the 95% confidence level across various fixed effects and control variables. However, since the core explanatory variable is the logarithm of a composite index, the original economic meanings of sub-indicators have been reflected through multiple layers of process, the interpretation of the coefficients should thus be limited to effect direction and significance, not simplified elasticity. The benchmark results preliminarily support the validity of Hypothesis 1, namely, that digitalization can enhance future regional GTI capabilities.

Table 4. Baseline regression results

	Dependent variable: <i>lnGtech</i>			
	(1)	(2)	(3)	(4)
<i>lnDI</i>	8.907*** (1.374)	2.741*** (0.978)	4.043*** (1.009)	2.683*** (1.004)
<i>lnIS</i>			0.017 (0.051)	-0.114** (0.056)
<i>lnGDP</i>			0.443*** (0.050)	0.126** (0.056)
<i>lnGovex</i>			0.128*** (0.035)	0.044 (0.036)
<i>lnFin</i>			0.230*** (0.037)	0.106*** (0.039)
<i>lnEdu</i>			-0.082** (0.040)	0.006 (0.042)
<i>lnElec</i>			0.317*** (0.061)	0.247*** (0.064)
Control	N	N	Y	Y
County FE	Y	Y	Y	Y
Year FE	N	Y	N	Y

R ²	0.491	0.561	0.540	0.566
N	5753	5753	5753	5753

Standard errors in parentheses, clustered at the county level, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

5.2. Robustness tests

5.2.1. Model misspecification

The estimation results from the former baseline model are prone to disturbance of modeling and omitted variables. This section tests these problems correspondingly. The most vulnerable aspect is the model's linear assumption. Therefore, we incorporated a quadratic term for the core explanatory variable to form a quadratic function of $\ln DI$. A significant coefficient for this quadratic term would indicate the presence of high-dimensional factors potentially affecting the estimation. The corresponding results are presented in the first two columns of Table 5. We find that the coefficient for the quadratic term of the core explanatory variable is insignificant and highly unstable. This suggests no evidence that the previous estimates were influenced by high-dimensional nonlinear factors, implying the reliability of the regression results in this regard.

The measurement approach for key variables may also introduce instability in coefficient estimates. We therefore altered the dependent variable measurement from total innovation to innovation efficiency—specifically, innovation per unit of GDP and per capita innovation. The regression results using these new variables are presented in the next four columns of Table 5. Changing the dependent variable measurement did not alter the direction or significance of the coefficients, while the coefficients themselves appropriately decreased, as expected.

Table 5. Robustness: Nonlinearity and innovation measurement

	<i>lnGtech</i>		<i>ln(Gtech/GDP)</i>		<i>ln(Gtech/population)</i>	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>lnDI</i>	4.034*** (1.008)	2.436** (1.016)	0.177** (0.081)	0.310*** (0.104)	0.645** (0.267)	0.012** (0.005)
<i>lnDI</i> ²	29.346 (75.919)	-160.728 (98.657)				
Control	Y	Y	N	Y	N	Y
County FE	Y	Y	Y	Y	Y	Y
Year FE	N	Y	Y	Y	Y	N
R ²	0.662	0.663	0.689	0.712	0.607	0.296
N	2457	2457	5753	5753	5753	5753

Standard errors in parentheses, clustered at the county level, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The variable *Gtech* used in this paper is the number of acquired green patent, which is typically nonnegative count data. Employing a Poisson regression model may be more effective and can avoid

estimation biases caused by taking the logarithm of discrete data (Cohn et al., 2022). Given the prevalence of zeros in the patent data, a zero-inflated negative binomial (ZINB) model is needed. The regression results are presented in the first two columns of Table 6, with incidence rate ratios (IRR) reported in square brackets. The IRR indicates the multiple by which the dependent variable changes relative to the baseline level when the independent variable increases by one unit. Regardless of whether the control variables are included, the coefficient of the digital lag term is significantly positive, and the IRR is consistently greater than one. This result further validates the robustness of the benchmark regression findings, which are not significantly affected by model specification choices.

Given that innovation activities are likely to depend on existing innovation capacity, this implies that innovation models may need to account for dynamic autocorrelation. In this section, we supplement our analysis with a dynamic autocorrelated panel model using the difference GMM estimation method, with the results presented in Table 6 column (3) and (4). The findings show that the first-order lag term for green technological innovation has a significant coefficient, whereas no clear evidence supports the second-order lag term. This result confirms the presence of first-order autocorrelation in green technological innovation. After controlling for autocorrelation, the positive effect of digitalization remains statistically significant, which is consistent with the benchmark regression. Therefore, we conclude that Hypothesis 1 is fundamentally robust.

If variables with significant effects on green technological innovation are omitted, the estimated results may be overstated. Conversely, if there are variables with significant effects on the independent variables, this may lead to endogeneity issues. To address these concerns, we strengthened the fixed effects in our model by controlling for annual characteristics at the provincial level across different EFZs, thereby accounting for common policies and other shocks at the provincial level. The results in the last two columns of Table 6 indicate that after incorporating enhanced fixed effects, the coefficient for *lnDI* remains consistent with the baseline regression and is significant at least at the 10% confidence level.

Table 6. Robustness: Count data, autocorrelation, and unobserved confounders

	ZINB: <i>Gtech</i>		GMM: <i>lnGtech</i>		HDFE: <i>lnGtech</i>	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>lnDI</i>	0.028** (0.014) [1.029]	0.079*** (0.013) [1.082]	29.344** (14.035)	32.220** (15.182)	1.697* (0.892)	1.713* (0.907)

<i>L.lnGtech</i>			0.441*** (0.144)	0.322* (0.178)		
<i>L2.lnGtech</i>				0.224 (0.177)		
Control	N	Y	N	Y	N	Y
County FE	Y	Y	Y	Y	Y	Y
Province#Year FE	N	N	N	N	Y	Y
AR(2)	-	-	0.016	0.626	-	-
Hansen test	-	-	0.100	0.151	-	-
R ²	-	-	-	-	0.634	0.635
N	5753	5753	5230	4707	5753	5753

Standard errors in parentheses, clustered at the county level, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; IRR in square brackets.

Following these targeted robustness checks, we have verified that the benchmark regression results are not significantly affected by model specification or omitted variable issues, rendering our conclusions reasonably reliable. However, endogeneity concerns remain to be addressed. Given the difficulty in constructing instruments at the county level, we proceed to re-estimate the results via a quasi-natural experiment based on digital policy shocks at the county level.

5.2.2. Quasi-natural experiment

As a rural digitalization initiative, the Digital Village pilot policy advances rural digital infrastructure and technological adaptation; empirical evidence confirms its multifaceted benefits including improved rural governance efficiency (He et al., 2025), agricultural industrial upgrading (Li et al., 2025), higher agricultural green total factor productivity (Cai & Han, 2024), boosted local entrepreneurship and factor inflow (Li & Tong, 2025), as well as tighter linkages with the national innovation system (Chen & Song, 2025). From the perspective of the NSE, Digital Village effectively activates the structural evolution, thus could cast positive effects on GTI.

The E-commerce in Rural Areas pilot policy provides dedicated funding to build rural e-commerce public service systems, logistics distribution networks, and internet penetration. Regarding on endowment structure, rural e-commerce contributes to the conversion of arable land non-grainization (Li et al., 2025) and pulls agriproducts selling prices (Liu et al., 2021), thus accelerate capital accumulation in rural areas. It also mitigates rural population outflow by creating local employment (Liu et al., 2025; Zhao et al., 2024), and stimulates rural women entrepreneurship (Dong et al., 2024). These accelerate the labor factor accumulation and upgrading. Lastly, the pilot policy promotes urban-rural commonwealth (Wei et al., 2024), creating a closer cooperation network. Together, these policies serve as plausible exogenous shocks to investigate the effect of digitalization on GTI.

Notably, these two pilot policies are already included in the baseline DI, this creates a concern of circular reasoning, which requires verifying that the presence of pilot policies in the index does not drive the baseline results. Accordingly, before conducting the DID design, we first reconstruct the digitalization index excluding all pilot policies. The updated entropy weights are reported in Appendix B. Baseline regressions using this cleaned index give consistent results, confirming the applicability of this DID design. We now exploit the plausibly exogenous variation from these digital policies as an alternative identification strategy. The estimations for the digital village pilot policy are presented in Table 7. The results indicate that the digitalization shock in infrastructure and commercial applications effectively enhance local GTI capabilities.

Table 7. Robustness: DID validation

	Dependent variable: <i>lnGtech</i>					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>did dv</i>	0.085* (0.044)	0.076* (0.042)	0.081** (0.037)			
<i>did_ecommerce</i>				0.188*** (0.036)	0.146*** (0.035)	0.067*** (0.026)
Control	N	Y	Y	N	Y	Y
Province FE	N	N	Y	N	N	Y
Year FE	N	N	Y	N	N	N
R ²	0.061	0.139	0.357	0.009	0.144	0.514
N	6276	6276	6276	6276	6276	6276

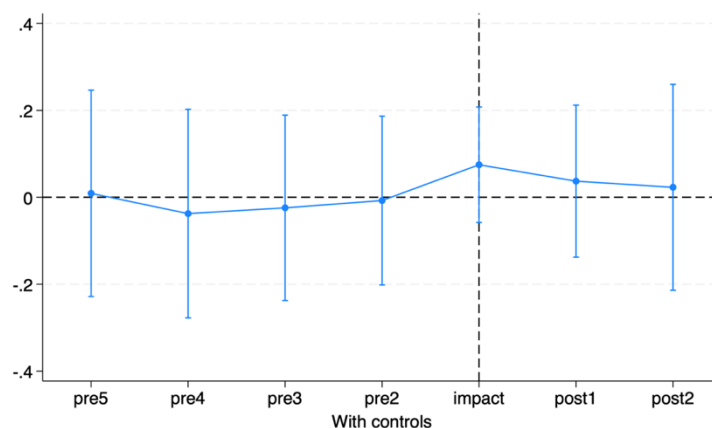
Standard errors in parentheses, clustered at the county level, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The applicability of quasi-experiments requires comparability across groups, which is known as the parallel trend assumption, so we investigate the generating process of the treated group. The selection of pilot areas for the Digital Village pilot areas was generated through a process of combining local recommendation with expert review, where provincial administrations recommend eligible counties or districts, and a joint review by multiple departments determine the final list². The selection of pilots for the E-Commerce in Rural Areas followed a very similar procedure. For quasi-experiments, the selection is not entirely random, the pilot areas may possess better conditions for digitalization, creating ex-ante differences. However, there are two reasons why these policies remain applicable: First, in China's administrative practices, it's common to balance development potential and development needs in the recommendation and approval process³, thereby mitigating pre-treatment differences; Second, DID studies using these two policies have repeatedly verified parallel trends with or without covariates (Li & Tong, 2025; Wei et al., 2024; Zhao et al., 2024), this

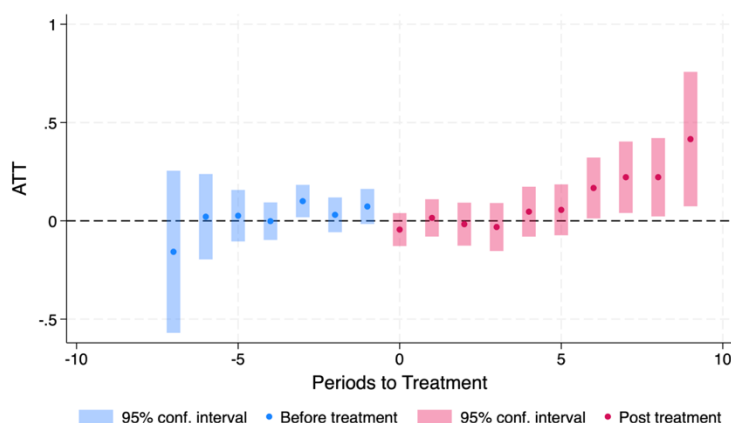
² https://www.cac.gov.cn/2020-09/18/c_1601988147662407.htm

³ https://www.cac.gov.cn/2020-07/17/c_1596539938841028.htm

paper also checks the parallel trends using simple event study method for Digital Village as shown in Figure 6a, since the pilot areas of E-Commerce in Rural Areas have several batches, the C-S estimator (Callaway & Sant'Anna, 2021) is adopted to account for heterogeneous treatment effects as in Figure 6b. These tests along with prevailing literature, provide justifications for the DID design in this section.



(a). Event study plot of Digital Village



(b). CS-DID event study plot of Rural E-commerceⁱ

Note: simple aggregated $\text{pre_avg.} = 0.013(\text{se} = 0.031)$, $\text{post_avg.} = 0.105^{**}(\text{se} = 0.053)$, control: never treated.

Figure 6. Parallel-trend tests of the DiD design

5.3. Mechanism tests

5.3.1. Digital-green synergy

According to the theoretical derivation, micro digitalization should enhance the degree of coordination between digital and green technological innovation, which then benefits long-term *GTI* capacity. The detailed regression results in Table 8 panel A support the first part of this derivation. Regardless of how the control variables and the fixed effects are adjusted, the coefficients of digitalization hold positive significance, providing powerful evidence that digitalization enhances digital-green synergy. Also, to ensure robustness of measurement, this section adopts the digital-

green synergy of acquired inventions (*DGSi*) and of all patents (*DGS_t*), and conducts natural log transformation (*lnDGS_i*, *lnDGS_t*) and IHS transformation (*ihs_DGS_i*, *ihs_DGS_t*) respectively, to obtain variables with four different measurement methods. All estimation results remain consistent.

Tests for the mediation effects are necessary to ensure the existence and effectiveness of the acting mechanisms. Table 8 panel B presents the results of combining Bootstrap sampling and the ZLC mediation test, which is based on the method specified by Zhao et al. (2010). The results clearly show that the indirect effect via digital–green synergy is real and robust, and Hypothesis 2 is thus validated.

Table 8. Mechanisms: digital–green synergy

Panel A	<i>lnDGS_i</i> (1a)	<i>lnDGS_t</i> (2a)	<i>ihs_DGS_i</i> (3a)	<i>ihs_DGS_t</i> (4a)
<i>lnDI</i>	5.667*** (1.787)	3.494*** (1.017)	0.335** (0.148)	0.339*** (0.118)
Control	Y	Y	Y	Y
FEs	Y	Y	Y	Y
R ²	0.364	0.380	0.554	0.685
N	5753	5753	5753	5753
Dependent variable: <i>lnGtech</i>				
Panel B	(1b)	(2b)	(3b)	(4b)
<i>lnDGS_i</i>	0.065*** (0.008)			
<i>lnDGS_t</i>		0.106*** (0.013)		
<i>ihs_DGS_i</i>			1.815*** (0.171)	
<i>ihs_DGS_t</i>				2.810*** (0.157)
<i>lnDI</i>	2.459*** (0.921)	2.338** (0.958)	2.223*** (0.846)	1.869** (0.743)
<i>Ind_Eff</i>	0.224** {0.107}	0.345*** {0.111}	0.460** {0.200}	0.814*** {0.247}
Control	Y	Y	Y	Y
FEs	Y	Y	Y	Y
R ²	0.574	0.575	0.582	0.605
N	5753	5753	5753	5753

County-level clustered robust standard errors in parentheses, bootstrap standard errors in curly braces, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

5.3.2. Technology supply market

Similar work has also been conducted on the technology supply market. Given that log-

transformation is involved in this step and that several new studies argue that regular log-transformation could produce unstable and vague estimations (Chen & Roth, 2024), we use both regular and IHS transformations to ensure robustness.

To comprehensively evaluate the effectiveness of the technology supply market, both the supply side and the demand side should be taken into consideration. In the Table 9, *CTC* represents the degree of interregional technological innovation cooperation, which reflects the production network of innovation. *Cite* stands for cross-country patent citations, which reflect the demand for technological innovation outcomes.

Table 9 clearly shows that both interregional technological innovation cooperation and citation benefit significantly from digitalization, which means that the entire technology market, including the supply and demand sides, is effectively stimulated. Combining the results from Table 8, the existence of these two mechanisms is confirmed. Thus, Hypothesis 3 is validated.

Table 9. Mechanisms: technology supply market

Panel A	<i>lnCTC</i>		<i>lnCTC_IHS</i>		<i>lnCite</i>		<i>lnCite_IHS</i>	
	(1a)	(2a)	(3a)	(4a)	(5a)	(6a)	(7a)	(8a)
<i>lnDI</i>	0.019** (0.008)	0.024*** (0.008)	0.023** (0.010)	0.030*** (0.011)	0.082*** (0.029)	0.078*** (0.025)	0.087*** (0.030)	0.081*** (0.027)
Control	N	Y	N	Y	N	Y	N	Y
FES	Y	Y	Y	Y	Y	Y	Y	Y
R ²	0.139	0.158	0.143	0.163	0.778	0.792	0.775	0.790
N	5753	5753	5753	5753	5753	5753	5753	5753
Dependent variable: <i>lnGtech</i>								
Panel B	(1b)	(2b)	(3b)	(4b)	(5b)	(6b)	(7b)	(8b)
<i>lnCTC</i>	0.199*** (0.017)	0.169*** (0.017)						
<i>lnCTC_IHS</i>			0.164*** (0.013)	0.139*** (0.013)				
<i>lnCite</i>					0.158*** (0.020)	0.141*** (0.019)		
<i>lnCite_IHS</i>							0.127*** (0.018)	0.111*** (0.017)
<i>lnDI</i>	0.043*** (0.010)	0.048*** (0.010)	0.043*** (0.010)	0.048*** (0.010)	0.012 (0.0213)	0.016 (0.021)	0.014 (0.022)	0.018 (0.021)
<i>Ind_Eff</i>	0.004** {0.002}	0.004*** {0.001}	0.004** {0.002}	0.004*** {0.002}	0.013*** {0.003}	0.011*** {0.002}	0.011*** {0.002}	0.009*** {0.002}
Control	N	Y	N	Y	N	Y	N	Y
FES	Y	Y	Y	Y	Y	Y	Y	Y
R ²	0.365	0.387	0.366	0.388	0.479	0.488	0.474	0.484

N	5753	5753	5753	5753	5753	5753	5753	5753
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County-level clustered robust standard errors in parentheses, bootstrap standard errors in curly braces, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

5.3.3. Robustness tests for the mechanisms analyses

Although *DGS* essentially reflects the degree of coupling and coordination between key digital innovation and green technological innovation, it inevitably incorporates components of green innovation. This may lead to mechanical correlation with the dependent variable. Specifically, the mechanical correlation mainly originates from two sources: the shared *Gtech* component, and simultaneous changes caused by other common influencing factors, as shown in Figure 7. This section conducts corresponding treatments to ensure the reliability of the mechanism inferences.

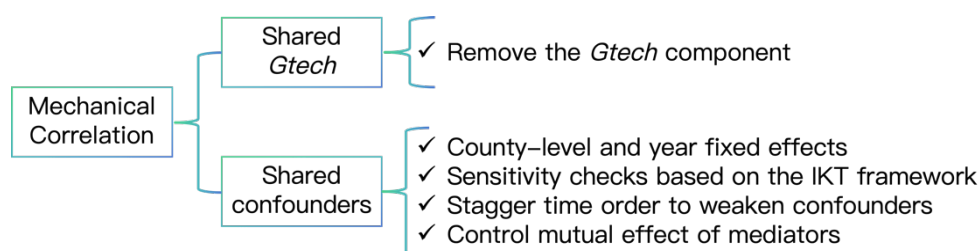


Figure 7. Design of the mediation robustness tests

First, the *DGS* with a shared *Gtech* component is replaced with *Dtech*, which has no overlap with the dependent variable *Gtech*, detailed results in Appendix Table C1 suggest that mechanical correlation doesn't drive the estimation solely. Second, based on the Imai-Keele-Tingley (IKT) mediation effect framework, this paper evaluates the sensitivity of mediation effects against confounders. According to the $ACME-\rho$ plots in Appendix Figure C1, the mediation effects are basically robust. Still, the causal mediation chains are staggered to furtherly weaken the simultaneity of potential confounders, and a joint effect model containing all three mediators is estimated. Both methodologies acquired consistent results supporting the robustness of the mediation effects.

5.4. Policy interaction

In addition to the business world, digitalization has popularized a series of digital infrastructures, hard/soft devices and corresponding management methods. Each of these could make a difference when applied in multiagent environmental regulation. For example, smart water quality detection devices and matched supervisory networks could lower the cost and enhance the efficiency of traditional water pollution regulation. In a broader range, activated vox populi on digital social platforms respond rapidly to public pollution and place greater pressure on governmental and private environmental regulation. On the other hand, stronger environmental regulation reinforces the

application of digital devices in economic green transition, thus enhancing the effect of digitalization on GTI. In conclusion, digitalization interacts with environmental regulation and thus increases GTI in a difficult way.

The results from Table 10 support the derivation; the interaction term of digitalization and environmental regulation is positively significant, as are the separate variables. This statistical result indicates that digitalization could strengthen the positive effect of environmental regulation on GTI.

Table 10. Environmental regulation interaction

	Dependent variable: <i>lnGtech</i>			
	(1)	(2)	(3)	(4)
<i>lnDI#lnER</i>	0.045*** (0.015)	0.030** (0.013)	0.032** (0.014)	0.029** (0.013)
<i>lnER</i>	0.183*** (0.040)	0.119*** (0.034)	0.117*** (0.037)	0.112*** (0.034)
<i>lnDI</i>	8.833*** (1.358)	3.060*** (0.985)	4.222*** (1.022)	2.992*** (1.019)
Control	N	N	Y	Y
County FE	Y	Y	Y	Y
Year FE	N	Y	N	Y
R ²	0.496	0.563	0.541	0.568
N	5753	5753	5753	5753

Standard errors in parentheses, clustered at the county level, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

5.5. Generalized theory validity

Innovation efforts in regions with differing levels of prior development may face distinct primary challenges. First, early-stage green innovation issues such as information asymmetry and knowledge barriers can be effectively mitigated through digitalization. However, once these issues are largely resolved, constraints on green innovation increasingly stem from core technological breakthroughs, institutional environments, and R&D talent—factors that cannot be advanced by digitalization alone. Second, as revealed by mechanism tests examining transmission channels, after coupled innovation based on digital–green synergy elevates green innovation levels, a divergence emerges between digital technology innovation and green technology innovation, creating tension that may weaken the positive correlation between digitalization and green technological innovation. Consequently, in advanced regions with strong green technological innovation capabilities, the driving effect of digitalization may diminish. Next, a panel quantile regression model was employed for verification.

The result is illustrated in Figure 8, a detailed regression table is provided in Appendix D.

Although the coefficient for digitalization levels decreases at higher green technology innovation quantiles, the driving effect of digitalization on GTI persists throughout, with a relatively gradual rate of diminishing returns. This phenomenon supports the aforementioned conjecture while revealing that digitalization's propulsive effect on regional GTI capabilities is stable across all quantiles, proving effective in lagging and advanced regions.

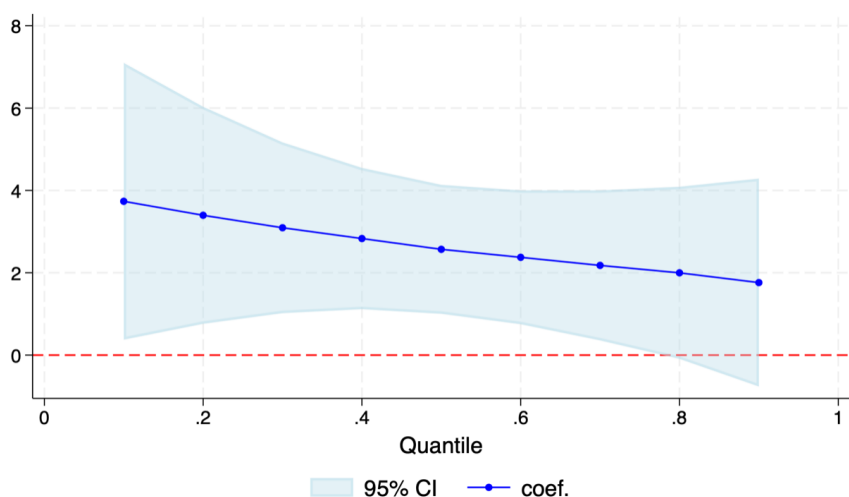


Figure 8. Quantile regression

5.6. Ecological endowment heterogeneity

The FMM model can uncover latent heterogeneous groups within the data. By incorporating covariates, it enables more accurate calculation of group probabilities. However, the number of heterogeneous groups cannot be predetermined, necessitating a comparison of information criteria across different grouping models to select the optimal number of groups. Considering the likelihood, degree of freedom, AIC and BIC, we specify two heterogeneous groups. Detailed statistics are provided in Appendix E.

Table 11 presents the final FMM results. The findings indicate two heterogeneous categories: for the majority of the sample, digitalization significantly promotes GTI, whereas for a minority, this effect is insignificant. The adjoint variables used to predict class membership are ecological service value (ESV) and the normalized difference vegetation index (NDVI). Although these two variables are measured through different approaches, one based on economic valuation and the other on satellite remote sensing, they serve as complementary proxies for the same latent construct: ecological endowment abundance.

The role of ecological endowment abundance in predicting the effectiveness of digitalization is

grounded in the mutual convergence framework of this study. On the supply side of mutual convergence, abundant ecological resources provide the comparative advantages that digitalization's network effect can leverage, serving as the fuel for structural adaptation. On the demand side, different ecological endowments generate distinct application scenarios for digital technologies, such as forest carbon monitoring, smart water management, and biodiversity tracking. Each scenario is viable only where the corresponding ecological resource exists, and together they shape the direction and intensity of the paradigm shift. In both directions, ecological endowment abundance conditions the effectiveness of mutual convergence. After ESV and NDVI were incorporated as adjoint variables, the class structure and coefficients remained largely stable, supporting the robustness of the classification. Furthermore, the coefficients of the adjoint variables align with theoretical expectations: the driving effect of digitalization on GTI is more pronounced in counties with higher ecological endowment abundance.

Table 11. Heterogeneity: Finite mixture model

	Dependent variable: <i>lnGtech</i>			
	(1)	(2)	(3)	(4)
<i>lnDI</i>	3.060** (1.379)	0.450 (1.904)	2.953*** (1.044)	0.528 (1.225)
<i>lnESV</i>			Base	-2.623* (1.499)
<i>lnNDVI</i>			Base	-7.234** (3.546)
Prob	0.773	0.227	0.776	0.224
Control	Y	Y	Y	Y
FEs	Y	Y	Y	Y
N	5753	5753	5753	5753

Standard errors in parentheses, clustered at the county level, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Thus, which districts and counties can achieve significant green innovation effects through digitalization? A more detailed identification requires the use of an interaction term heterogeneity model. China's key ecological functional zones are categorized into different types on the basis of their distinct ecological endowments and environmental characteristics. In this section, dummy variables are generated for each ecological functional zone type, and the significance of the interaction term coefficients is tested. $\beta_1 + \beta_3$ indicates the marginal effects across categories.

The results in Table 12 reveal that the impact of digitalization on green technological innovation is more pronounced in forestry, water conservation, and fragile ecological functional zones. These

zones harbor abundant ecological products (forestry economy, tropical fruits, ecological animal husbandry, and tourism), and the management and dynamic monitoring of forests and water sources are better suited to new digital technologies, yielding greater marginal benefits from digital applications for green innovation. Conversely, the effect is weaker and marginally insignificant in ecological functional zones focused on biodiversity conservation, soil and water conservation, and windbreak and sand fixation. This may stem from these areas' greater reliance on long-term engineering or biological interventions, where digital technologies offer relatively limited adaptability and immediate impact.

Table 12. Heterogeneity: EFZ type difference

	Dependent variable: <i>lnGtech</i>					
	Forest	Diversity	Water	Fixation	Prevention	Fragile
<i>lnDI#Type</i>	4.140** (2.099)	-4.965** (2.268)	5.191*** (1.802)	-4.277* (2.207)	-4.637** (2.090)	4.143** (2.089)
<i>lnDI</i>	2.177** (1.041)	3.373*** (1.134)	0.383 (1.146)	3.244*** (1.018)	3.435*** (1.080)	2.166** (1.042)
$\beta_1 + \beta_3$	6.317*** (1.904)	-1.592 (2.063)	5.574*** (1.477)	-1.034 (2.029)	-1.202 (1.858)	6.309*** (1.896)
Controls	Y	Y	Y	Y	Y	Y
FEs	Y	Y	Y	Y	Y	Y
R ²	0.567	0.567	0.568	0.567	0.567	0.567
N	5753	5753	5753	5753	5753	5753

Standard errors in parentheses, clustered at the county level, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The interaction term model yields results consistent with those of the FMM model, indicating that the analysis in this section aligns with economic realities and is robust. Heterogeneity tests reveal additional factors influencing the digital–green nexus: ecological endowments, represented by ecosystem service values and vegetation coverage indices, serve as key drivers for digitalization-enabled green technological innovation.

6. Conclusions and policy implications

Within an ecological economic system that exhibits substantial demand for green technologies, digitalization fulfills this role by driving digital-driven green productivity, thereby empowering green technological innovation in less developed, ecologically constrained economies like China's Ecological Function Zones. This research constructs a comprehensive county-level digitalization index to examine this conceptual linkage from a technology supply perspective, estimating its multidimensional effect on green technological innovation and exploring the insightful

heterogeneity within the established framework.

Causal inferences in this work suggest that micro digitalization induces significant improvement in GTI. This result remains robust after the model specification, omitted variables and endogeneity issues are checked. The digital–green synergy market and effective technology supply market serve as two powerful mechanisms. Digitalization also interacts with policies and promotes the effectiveness and efficiency of environmental regulation. Interestingly, although a diminishing marginal effect is observed, the positive effect holds true in all quantiles of GTI, which indicates the relative universality of the results. We use FMM to detect heterogeneity, and the information criteria propose 3 heterogeneous paths, with local ecological endowment affecting the path possibility. The positive effect of digitalization on GTI is stronger in areas with better ecological conditions, which include forest, water and fragile EFZs. These conclusions provide insights for the digital–green cooperation development of all later-comer economies, especially rural ecological areas.

First, the baseline results confirm that county-level digitalization has a significant and robust positive effect on GTI across all EFZs, and this effect persists even at lower levels of GTI. This implies that continued investment in county-level digital infrastructure, including broadband access, digital government services, and e-commerce platforms, constitutes a feasible entry point for promoting green innovation in these ecologically constrained regions. Given that EFZ counties typically face limited fiscal capacity and thin local tax bases, such investments would likely require support from vertical ecological compensation transfers rather than relying solely on local budgets.

Second, the mediation analysis identifies two distinct transmission channels that can inform county-level policy design. On the one hand, local governments can integrate digital development plans with green innovation targets, for example by embedding green technology indicators into county-level digital economy roadmaps. On the other hand, policies should facilitate participation in existing provincial-level technology transfer networks through subsidized digital matching services, rather than assuming that building local patent exchanges will generate sufficient activity in regions with thin local technology demand.

Third, the policy coordination analysis shows that digitalization amplifies the effectiveness of environmental regulation in promoting GTI. Digital tools such as remote sensing based monitoring and online compliance platforms can reduce enforcement costs at the county level. However, since the marginal effect of digitalization diminishes as the innovation base improves, counties with

stronger innovation foundations should shift focus from basic infrastructure to targeted digital applications that address specific local bottlenecks.

Finally, the heterogeneity analysis by EFZ type provides actionable guidance for differentiated policy design. In forest conservation, water conservation, and fragile EFZs, where digitalization yields stronger effects, policies should prioritize digital platforms for forest carbon sink monitoring, smart water management systems, and eco-tourism promotion. These interventions build directly on the existing ecological endowment advantages of these regions. In biodiversity conservation and wind-sand EFZs, where the direct effect is not significant, policies should focus on enabling functions such as digital biodiversity monitoring, ecological early warning systems, and cross-regional technology cooperation, instead of ecological value realization. This allows these areas to participate in green innovation value chains without requiring locally concentrated innovation capacity.

Some flaws exist due to data availability and other problems. The index system used in this study contains several macro-level variables due to difficulties in acquiring detailed county-level data. This flaw might blur the accuracy of the conclusions. Future researchers could discover more representative methods to reflect county-level characteristics to acquire more precise insights. Also, although this paper discusses the necessity of a log-linear model, taking the log of a composite index still makes it harder to interpret its economic meaning. This is a compromise balancing quantitative performance with econometric interpretation, which could benefit from deeper supplementary qualitative analyses. Lastly, due to several difficulties in finding more exogenous mediators under current variable definitions, the mechanical correlation and confounders are not eliminated completely. Thus the mediation analyses should be limited to correlational path inference, instead of causal claims. Further structural modelling might help address this issue.

Author Contributions Statement

Statement of conflict interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability statement

The data will be made available from the authors upon reasonable request.

Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

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Appendix

A. Functional form and the log-transformation of *DI*

This appendix section discusses the necessity and applicability of log-linear models in the context of this paper's research topic.

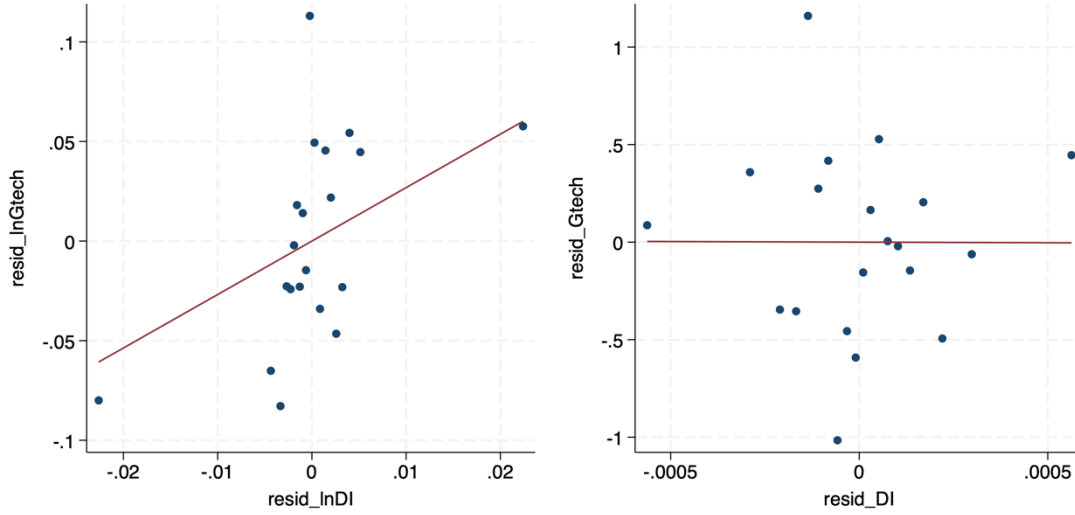
Given that classic models in the field of innovation mainly employ Cobb-Douglas production functions with exponential parameters, a strictly horizontal linear model may not be able to fully capture the effects of innovation determinants, which justifies the use of a logarithmic model. This section first examines this necessity by estimating panel regression models using the level values of *Gtech*, *DI*, and control variables; the suffix “_mm” indicates that the data have been normalized. As shown in Table A1, none of the level models yield significant estimates, and the coefficient estimates are highly unstable.

Table A1. The necessity of logarithmic function form

	<i>Gtech</i>		<i>Gtech_mm</i>	
	(1)	(2)	(3)	(4)
<i>DI</i>	40.175 (928.113)	-15.658 (873.482)		
<i>DI_mm</i>			0.131 (3.022)	-0.051 (2.844)
Control	Y	Y	Y	Y
FEs	Y	Y	Y	Y
R ²	0.594	0.597	0.594	0.597
N	5753	5753	5753	5753

Standard errors in parentheses, clustered at the county level, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

To compare the explanatory power of the logarithmic and linear models, we plotted Binscatter for both models. As shown in Figure A1, the pattern of correlation between green innovation performance and digitalization is only evident in the logarithmic model; therefore, it remains necessary to use the logarithmic-linear model.



(a). Binscatter plot of log-function

(b). Binscatter plot of level function

Figure A1. Binscatter plot of different function forms

However, the core issue is that the standard logarithm does not behave naturally near 0 (Chen & Roth, 2024), while in the dataset used in this paper, the minimum value of DI is 0.002, which is very close to 0. Therefore, there may be an issue with numerical disperse after transformation. We consider using the inverse hyperbolic sine transformation, as shown in Equation I, to address this.

$$y^{IHS} = \ln \left(y + \sqrt{y^2 + 1} \right) \quad (I)$$

Yet further analysis in Table A2 reveals that the DI after the IHS transformation is almost identical to the level value and is also highly similar to the natural logarithm of the level value plus 1; only the natural logarithm transformation maintains an appropriate functional difference from the level value. This is because the purpose of the IHS transformation is to make the transformed variable's values closely approximate the original values at low levels; since the range of DI in Table A2 is [0.002, 0.556], which is generally within a small range, the IHS transformation closely approximates the level value. Since DI does not actually contain zero values, this paper does not adopt the $\ln(x+c)$ transformation method. Nevertheless, the comparison in Table 6 shows that $\ln(DI+1)$ is extremely close to $\ln DI$. Performing a Taylor expansion of $\ln(x+1)$ yields:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \quad (II)$$

Which tells that, when the value of x is small enough, we have $\ln(1+x) \approx x$. This means that to effectively perform a log-linear transformation on DI , the best choice is the natural logarithm transformation.

Table A2. Correlation matrix of DI in different transformation forms

	Mean	Median	SD	Min	Max	Skewness	Kurtosis	Corr
<i>DI</i>	0.137	0.117	0.092	0.002	0.556	0.943	4.109	1.000
<i>lnDI</i>	-2.364	-2.145	1.128	-6.109	-0.587	-1.745	5.717	0.805***
<i>ln(DI+1)</i>	0.125	0.111	0.079	0.002	0.442	0.702	3.425	0.998***
<i>IHS_DI</i>	0.136	0.117	0.090	0.002	0.531	0.883	3.883	0.999***

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

As for the issue of value dispersion, we firstly check the sample observations of *DI* near 0, and plotted the cumulative distribution of *DI*'s level value. As shown in Figure A2, there are only 6.4% of sample near 0, therefore is unlikely to cause significant biases on general estimations.

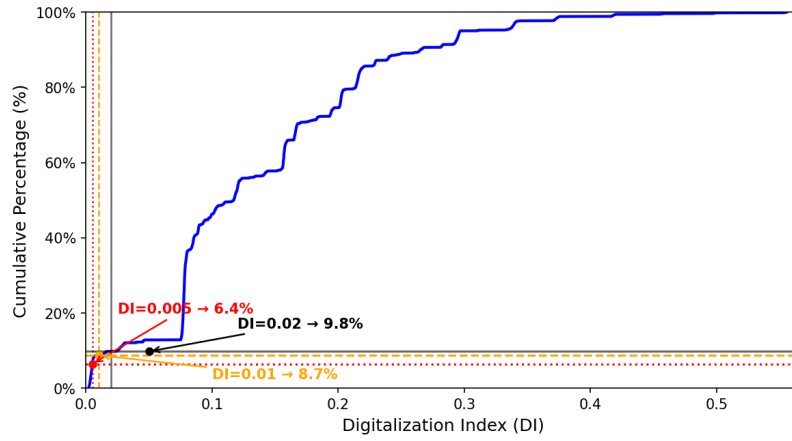


Figure A2. Cumulative distribution plot of *DI*

To evaluate the influence of post-log outliers on estimations, we present a partial scatter plot of *lnGtech* and *lnDI*. As shown in Figure A3, even beyond the outliers, a large number of observations still exhibit a clear linear pattern, which means that the observations with excessively dispersed low values do not dominate the estimation.

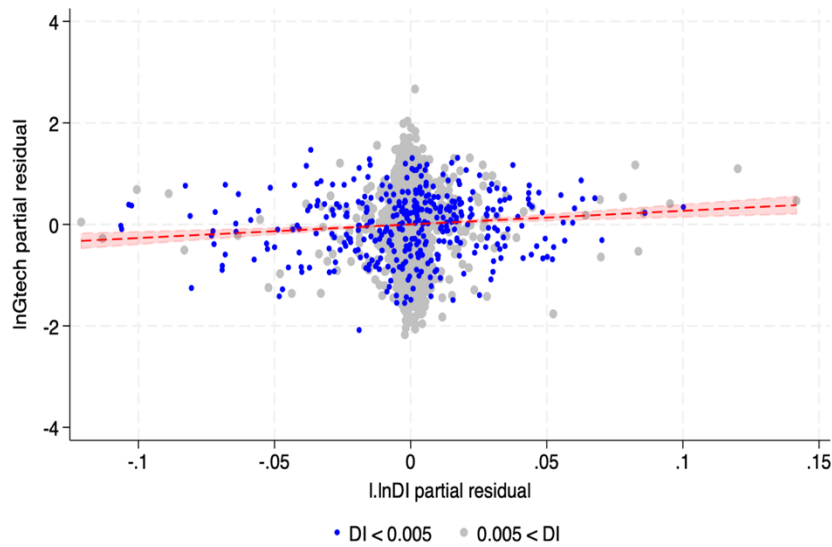


Figure A3. Partial scatter plot of *lnGtech* and *lnDI*

Table A4 presents the regression results after excluding samples with $DI < 0.005$, which indicate that the baseline estimations are not driven by tail samples. Also, we generate a dummy variable *Tail* denoting samples with $DI < 0.005$, and conduct interaction regressions to examine cross-group differences. As shown in column (5) and (6) of Table A4, the interaction terms are not significant, and the coefficients themselves are adequately small, suggesting no meaningful difference between tail and other samples.

Table A4. Excluding samples with DI near zero

	Dependent variable: $\ln Gtech$					
	(1)	(2)	(3)	(4)	(5)	(6)
$\ln DI$	12.481*** (3.277)	3.495*** (1.136)	5.633*** (1.589)	3.641*** (1.289)	2.721*** (1.002)	2.662** (1.031)
$Tail \times \ln DI$					0.004 (0.020)	0.004 (0.018)
Control	N	N	Y	Y	N	Y
County FE	Y	Y	Y	Y	Y	Y
Year FE	N	Y	N	Y	Y	Y
R^2	0.497	0.569	0.549	0.575	0.561	0.566
N	5386	5386	5386	5386	5753	5753

Standard errors in parentheses, clustered at the county level, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Finally, this part verifies from multiple perspectives that, given this paper's research topic and data situation, log-linear models are required, and natural log-transformation of DI is optimal according to its distribution. Also the dispersion issue of log-transformation near 0 doesn't significantly affect the robustness of baseline results.

B. Circular validation and comparability of the DiD design

Appendix Table B1 provides updated index system excluding pilot policies and the entropy weights of each sub-indicators.

Table B1. Excluding policy dummy-variables from DI

Sub-indicator	Original weight	New weight
Digital inclusive finance	0.009226	0.016806
Total volume of postal and telecommunications services	0.263846	0.297009
Proportion of employees in industries	0.055277	0.077901
Number of fixed-line telephone subscribers	0.074903	0.186326
Number of mobile phone subscribers	0.108377	0.280576
Number of broadband internet access users	0.084741	0.141384

Table B2 presents the baseline regression results using this cleaned index ($\ln DI_{clean}$). The coefficients remain positive and statistically significant across all specifications, with magnitudes

comparable to those in Table 3. This consistency confirms that the exclusion of policy indicators does not alter the baseline conclusion, thereby removing the circularity concern before proceeding to the DID estimation.

Table B2. Recheck of baseline regressions

	Dependent variable: <i>lnGtech</i>			
	(1)	(2)	(3)	(4)
<i>lnDI clean</i>	9.026*** (1.439)	3.728*** (1.128)	2.870** (1.164)	3.276*** (1.153)
Control	N	N	Y	Y
County FE	Y	Y	Y	Y
Year FE	N	Y	N	Y
R ²	0.490	0.561	0.538	0.567
N	5753	5753	5753	5753

Standard errors in parentheses, clustered at the county level, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

C. Robustness tests for the mechanism analyses

To conduct a robustness test for this mechanism, this section directly adopts the core digital innovation component of the *DGS* indicator as the mediating variable. First, digital innovation and green innovation are non-overlapping, and factors affecting general innovation have been controlled as covariates. There are unlikely to exist shared influencing factors for digital and green innovation beyond the control variables, which rules out obvious mechanical correlation. Second, if digitalization can genuinely advance digital-green synergy, digital innovation will progress alongside improvements in green innovation, which should manifest as a positive correlation between the two. Accordingly, digital innovation itself can serve as a mediator to verify the aforementioned mechanism. Table C1 presents the test results using *Dtech*, all of which show significantly positive values, consistent with the inferences drawn from the results in Table 8.

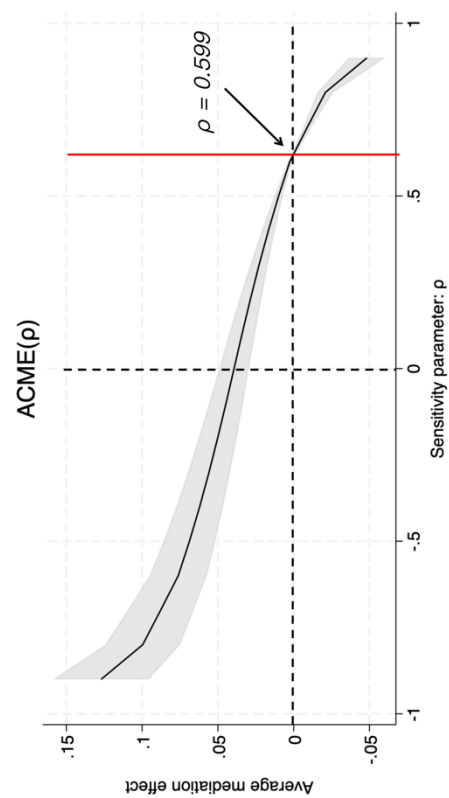
Table C1. Mechanism robustness: replace *lnDGS*

	<i>lnDtech</i>	<i>ihs_Dtech</i>	<i>lnGtech</i>	<i>lnGtech</i>
	(1)	(2)	(3)	(4)
<i>lnDI</i>	3.591*** (1.281)	4.225*** (1.483)	1.791** (0.887)	1.826** (0.909)
<i>lnDtech</i>			0.259*** (0.017)	
<i>ihs_Dtech</i>				0.206*** (0.014)
<i>Ind_Eff</i>			0.891*** {0.234}	0.857*** {0.219}
Control	Y	Y	Y	Y
FEs	Y	Y	Y	Y
R ²	0.659	0.630	0.599	0.596
N	5753	5753	5753	5753

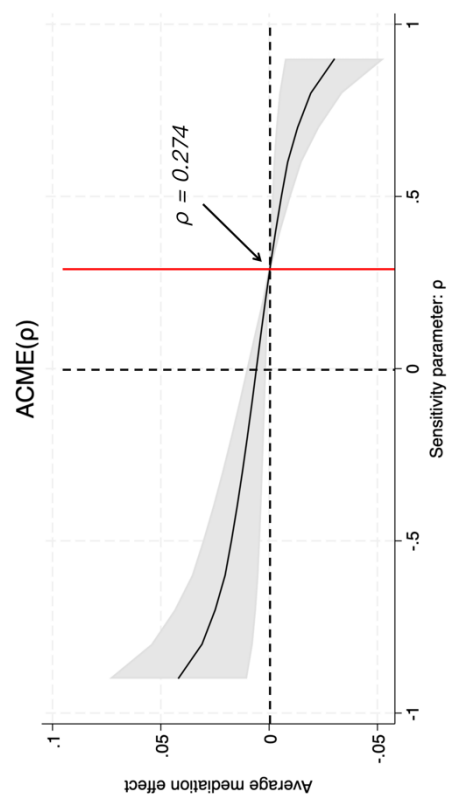
County-level clustered robust standard errors in parentheses, bootstrap standard errors in curly braces, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Although using *Dtech* directly for regression could remove a large portion of the mechanical correlation between *DGS* and *Gtech*, these two innovation performances might yet be influenced by other same variables, such as institutional quality or human capital. In the mediation analyses, our control variables and county-level fixed effect already largely covered these factors, partial out observable and unobservable common influencing factors, this paper still conducts further test to eliminate endogeneity caused by potential omitted variables. We first conduct sensitivity tests based on the IKT framework on the mediation effects' propensity of being confounded by omitted variables.

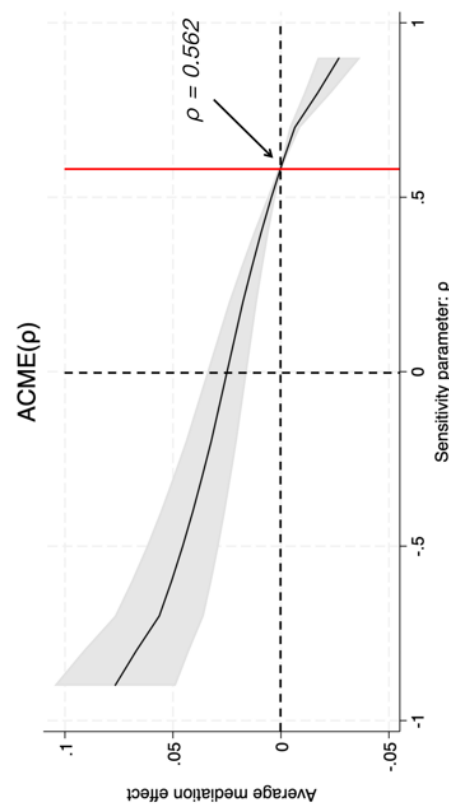
Figure C1 presents the relationship between the average causal mediation effect (ACME) and the sensitivity parameter ρ , detailed calculation and estimation methods are in accordance with the original literature from Imai et al. (2010). The value range of $|\rho|$ is $[0, 1]$, with larger absolute value suggesting the ACME is harder to be affected by confounders. Normally, $|\rho| \geq 0.3$ is taken as a reference threshold for robust ACME. The four sub-figures in Figure 11 each contain a red vertical line denoting the ρ value when $ACME = 0$. It is supported that digital-green synergy and technology market optimization basically satisfy the robustness criteria of the IKT framework, only the ρ of *CTC* is relatively small, suggesting it's prone to potential confounders, which calls for further tests.



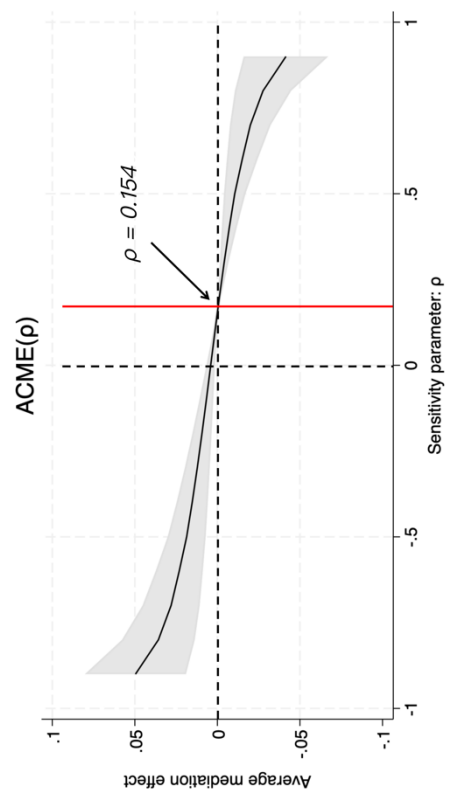
(b). Sensitivity plot of Diech



(d). Sensitivity plot of Cite



(a). Sensitivity plot of DGS



(c). Sensitivity plot of CTC

Figure C1. The mediation paths' sensitivity against confounders

Staggering the independent, dependent and mediating variables on the time order could help weaken the simultaneity of potential confounders, and also clarify effect order, preventing mutual disturbing between independent variables and the mediators, while excluding endogeneity caused by reverse causality. We lag $\ln DI$ for 1 period, place it on the $t-1$ time point, the mediators and control variables are placed on t , while $\ln Gtech$ is forwarded to $t+1$. Table C2 provides corresponding regression results: the coefficients of $\ln DI$ and the mediators remain positively significant. Although the effectiveness of this method shouldn't be exaggerated, these results at least suggest that recursive causality and other confounders in each period are not able to solely explain the correlation between $\ln Gtech$, the mediators and $\ln DI$.

Table C2. Mechanism robustness: staggering time points to weaken shared confounders

	$\ln DGSi$	$F.\ln Gtech$	$\ln cite$	$F.\ln Gtech$	$\ln CTC$	$F.\ln Gtech$
	(1)	(2)	(3)	(4)	(5)	(6)
$L.\ln DI$	5.667*** (1.787)	2.257** (1.108)	0.032* (0.018)	0.054*** (0.012)	0.024*** (0.008)	0.057*** (0.013)
$\ln DGSi$		0.051*** (0.008)				
$\ln Cite$				0.173*** (0.014)		
$\ln CTC$						0.137*** (0.022)
Control	Y	Y	Y	Y	Y	Y
FES	Y	Y	Y	Y	Y	Y
R ²	0.364	0.596	0.683	0.425	0.158	0.393
N	5753	5230	5753	5230	5753	5230

Robust standard errors in parentheses, clustered on province-year, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Since multiple mechanisms are proposed, the confounders of the CTC path is thus likely to be other proven mediators. So we run a joint effect test containing all three mediators simultaneously, and substitute $\ln DGS$ with $\ln Dtech$, which has much less mechanical correlation with the dependent variable $\ln Gtech$. This joint effect model investigate if the mediation effects survive after controlling each other paths. Results in Table C3 suggest that, after excluding the influence of mutual confounding in the mediators, the formerly proposed paths still hold.

Table C3. Mechanism robustness: joint effect tests

	$\ln Gtech$			
	(1)	(2)	(3)	(4)
$L.\ln DI$	0.037*** (0.010)	0.040*** (0.011)	0.020** (0.010)	0.024** (0.010)

<i>L.lnDtech</i>	0.483*** (0.017)	0.470*** (0.016)	0.381*** (0.015)	0.377*** (0.015)
<i>L.lnCTC</i>	0.076*** (0.022)	0.069*** (0.021)	0.063*** (0.019)	0.057*** (0.019)
<i>L.lnCite</i>	0.066*** (0.009)	0.063*** (0.009)	0.101*** (0.009)	0.087*** (0.009)
Control	N	Y	N	Y
FEs	N	N	Y	Y
R ²	0.436	0.441	0.504	0.507
N	5753	5753	5753	5753

Robust standard errors in parentheses, clustered on province-year, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Nevertheless, it must be acknowledged that the mechanical correlation between key digital innovation and green technological innovation cannot be completely removed under current variable construction, unless better measurements or quantitative methodologies are available. Meanwhile, since the mediation paths this research endeavors to discuss are the synergy between two types of innovation, and the collaboration of innovative activities, the search for more exogenous variables is beyond complex. This paper tries to split innovation categories, and control common influencing factors of innovation via closely related covariates and county-level FE, also these robustness tests at least strengthen the credibility and persuasion of the mediation inference, still the validity boundaries of such inference must be emphasized.

D. Detailed quantile regression results

The regression results table presented below is consistent with Figure 8 in the main text.

Table D1. Quantile regression results

Quantile	Coef.	Std. err.	z	P> z	[95% Confidence Interval]	
0.1	3.736	1.715	2.18	0.029	0.375	7.098
0.2	3.396	1.345	2.53	0.012	0.760	6.0336
0.3	3.094	1.058	2.92	0.003	1.019	5.168
0.4	2.832	0.876	3.23	0.001	1.115	4.548
0.5	2.570	0.799	3.22	0.001	1.004	4.136
0.6	2.376	0.829	2.86	0.004	0.750	4.002
0.7	2.179	0.930	2.34	0.019	0.357	4.002
0.8	1.999	1.067	1.87	0.061	-0.092	4.089
0.9	1.762	1.288	1.37	0.172	-0.764	4.287

E. FMM grouping statistics

Table E1 presents the relevant statistics for determining the optimal number of heterogeneous groups in the FMM. We observe a disagreement between AIC and BIC here: AIC is minimized at 3 groups, whereas BIC is minimized at 2 groups. Given that AIC places greater emphasis on predictive

accuracy and model generalizability, while BIC exhibits stronger consistency and a preference for interpretability, we prioritize BIC and specify 2 heterogeneous groups.

Table E1. Heterogeneity: Finite mixture model, panel A

Class number	N	Likelihood	df	AIC	BIC
No adjoint variables					
1	5753	-4957.484	19	9952.967	10079.46
2	5753	-4839.435	39	9756.870	10016.51
3	5753	-4779.954	59	9677.908	10070.70
With adjoint variables					
1	5753	-4957.484	19	9952.967	10079.46
2	5753	-4836.048	41	9754.097	10027.05
3	5753	-4777.575	63	9681.150	10100.57

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